HW05 — AI-assisted Performance Testing

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Repository: https://github.com/linhnph05/hw05

1. Purpose

This report explains how I designed, ran, and reviewed performance tests for the EShop backend. I used Apache JMeter for three scenarios: Load, Stress, and Spike. I also ran a ten-minute endurance test. An AI assistant helped with the first design and result analysis, but I checked the plans and raw results myself.

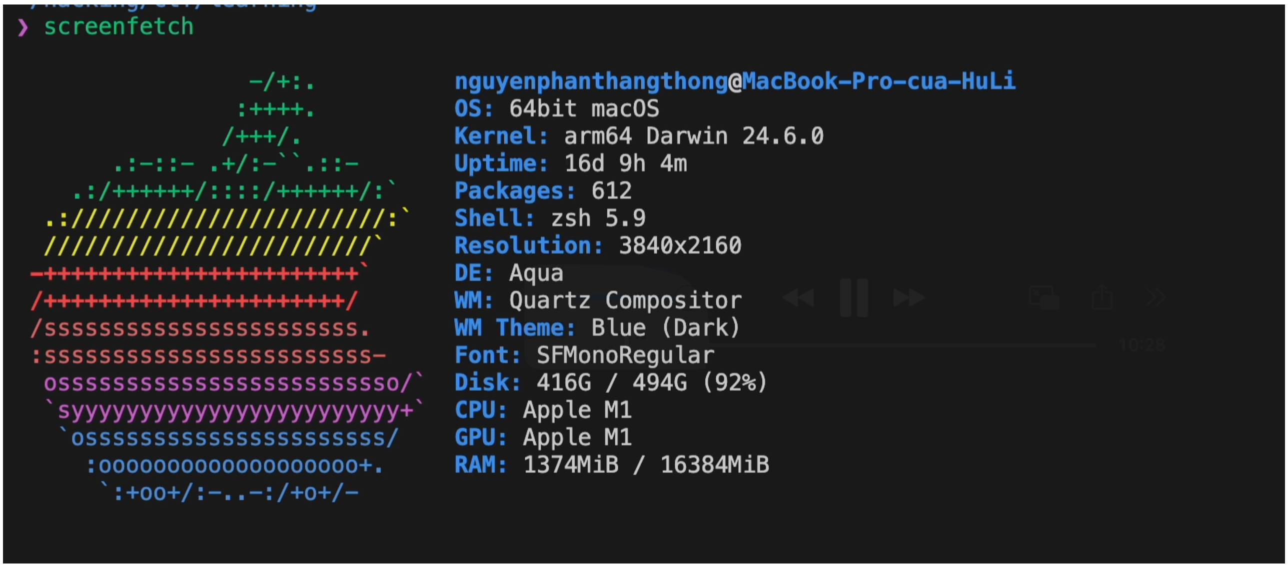
The report follows the same order as my work:

1. choose the endpoint scope;
2. review the AI design;
3. prepare the final JMeter plans and test data;
4. run the tests and monitor the backend;
5. review the measured results;
6. check the AI analysis; and
7. propose continuous performance testing.

2. System and test environment

Item	Value
System under test	EShop Node.js, Express, and SQLite backend
Base URL	`http://127.0.0.1:3000`
Load-testing tool	Apache JMeter 5.6.3
Test mode	JMeter non-GUI mode
Live resource evidence	`htop`, filtered to the backend PID
Numeric resource data	macOS `ps`, sampled once per second
Hardware evidence	`screenfetch`
Computer	MacBook Pro (MacBookPro17,1), Apple M1, 8 CPU cores, 16 GB RAM
Operating system	macOS 15.7.3
AI assistant	Codex / GPT-5.6

The following `screenfetch` screenshot records the test computer and hostname. It shows the Apple M1 CPU and GPU, 16 GB RAM, macOS, and the display resolution.



I used JMeter CLI mode because the official JMeter Getting Started guide says that load testing must run without the GUI for optimal results. The guide is at https://jmeter.apache.org/usermanual/get-started.html#non_gui. This also prevents the JMeter GUI from using resources that could affect the test.

3. Step 1 — Choose the test scope

HW05 requires three endpoint groups: read-heavy, auth-heavy, and transactional. I used one group for each test type. I chose a workflow of three endpoints inside each group.

Test	Endpoint group	Workflow
Load	Read-heavy	`GET /api/cart` → `GET /api/orders/my-orders` → `GET /api/orders/{id}`
Stress	Auth-heavy	`POST /api/register` → `POST /api/forgot-password` → `POST /api/reset-password`
Spike	Transactional	`POST /api/apply-coupon` → `POST /api/admin/coupons` → `DELETE /api/admin/coupons/{id}`

The Load workflow contains only read requests. The Stress workflow performs a complete password life cycle. It does not use login, so the three-failed-login lockout is not triggered. The Spike workflow applies a coupon, creates a temporary coupon, and deletes that coupon.

4. Step 2 — Review the AI test design

I first asked the AI to suggest endpoints, user counts, ramp-up times, test durations, and report views. The first answer used one endpoint in each scenario. This meets the endpoint-group rule, but I chose a broader workflow with three endpoints in each scenario.

The AI also suggested up to 50 users for the Stress test. I started with 30 users because the test writes to a local SQLite database on my laptop. I used smoke tests before the measured runs.

My final changes to the AI design were:

- add a separate CSV file for every scenario;
- add HTTP status assertions;
- extract the newest order ID in the Load workflow;
- extract the password-reset token in the Stress workflow;

- extract and delete the temporary coupon in the Spike workflow;
- use three different JMeter listener types; and
- reduce the first Stress run from 50 users to 30 users.

5. Step 3 — Prepare the final test plans

Test plan	CSV input	JMeter report view	Main parameters
`23127081_Load_20260816.jmx`	`test-data/read_input.csv`	Summary Report	20 users, 60-second ramp-up, 300 seconds, 1-second think time
`23127081_Stress_20260816.jmx`	`test-data/auth_input.csv`	Aggregate Report	30 users, 60-second ramp-up, 180 seconds, 500 ms think time
`23127081_Spike_20260816.jmx`	`test-data/transaction_input.csv`	View Results Tree	2 → 30 → 2 users, three 60-second stages

Each plan reads its own CSV file. The three listener types are different, as required. For the real measurements, the raw JTL and generated HTML dashboard are the main evidence.

The Load plan has a one-user setup group that creates one order before the measured traffic starts. The setup sample is excluded from the JMeter results, so it does not change the three measured endpoints. The plan then reads `/api/orders/my-orders`, extracts the newest order ID, and uses that value for `GET /api/orders/{id}`. The endurance run reuses the same setup.

6. Step 4 — Run the tests

Before every run, I started the EShop backend, found the backend PID, and started one-second CPU and RSS sampling. I then ran JMeter from the repository root. The important options are:

- `-n`: run without the JMeter GUI;
- `-t`: select the JMeter plan;
- `-l`: save the raw JTL file;
- `-j`: save the JMeter log; and
- `-e -o`: generate the HTML report.

6.1 Load command

```
jmeter -n -t test-plans/23127081_Load_20260816.jmx -l results/load.jtl -e -o results/load_html
-j results/load_jmeter.log
```

6.2 Stress command

```
jmeter -n -t test-plans/23127081_Stress_20260816.jmx -l results/stress.jtl -e -o
results/stress_html -j results/stress_jmeter.log
```

6.3 Spike commands

The Spike test has baseline, high-load, and recovery stages.

```
jmeter -n -t test-plans/23127081_Spike_20260816.jmx -l results/spike_baseline.jtl -e -o
results/spike_baseline_html -j results/spike_baseline_jmeter.log -Jspike_threads=2 -
Jspike_ramp=1 -Jspike_stage_duration=60
jmeter -n -t test-plans/23127081_Spike_20260816.jmx -l results/spike.jtl -e -o
results/spike_html -j results/spike_jmeter.log -Jspike_threads=30 -Jspike_ramp=5 -
Jspike_stage_duration=60
jmeter -n -t test-plans/23127081_Spike_20260816.jmx -l results/spike_recovery.jtl -e -o
results/spike_recovery_html -j results/spike_recovery_jmeter.log -Jspike_threads=2 -
Jspike_ramp=1 -Jspike_stage_duration=60
```

I combined the three stage results in time order for the submitted `results/spike.jtl`. The stage summary is in `results/spike_execution.md`.

6.4 Endurance command

I reused the Load workflow with 100 users for 600 seconds.

```
jmeter -n -t test-plans/23127081_Load_20260816.jmx -l results/endurance.jtl -e -o
results/endurance_html -j results/endurance_jmeter.log -Jload_threads=100 -Jload_ramp=60 -
Jload_duration=600
```

7. Step 5 — Review the measured results

7.1 Response results

Run	Samples	Throughput	Mean	p95	Maximum	Errors
Load	5,348	17.8 RPS	8.56 ms	30 ms	178 ms	0.00%
Stress	8,522	47.3 RPS	25.06 ms	139 ms	648 ms	0.00%
Spike baseline	230	3.83 RPS	10.43 ms	36 ms	139 ms	0.00%
Spike high load	3,255	54.25 RPS	23.37 ms	103 ms	622 ms	0.00%
Spike recovery	226	3.77 RPS	14.84 ms	40 ms	377 ms	0.00%

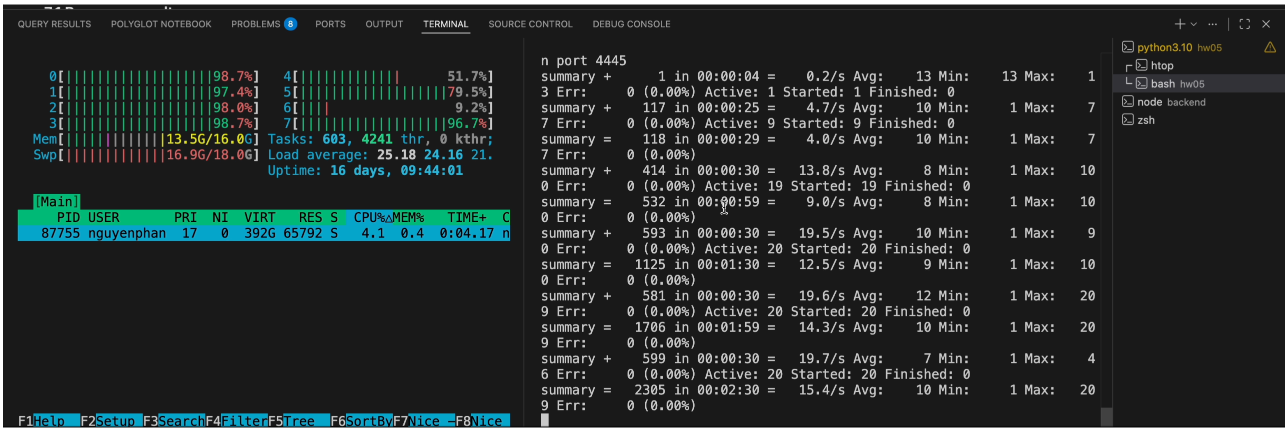
The Load test was stable at its planned traffic. The Stress test had a higher p95 than Load because it performed database writes, but it still completed without errors. During the Spike test, p95 increased to 103 ms at 30 users and returned to 40 ms in recovery. This shows that the backend recovered after the short traffic increase.

7.2 Backend resource results

Run	Average CPU	Peak CPU	Average RSS	Peak RSS
Load	1.99%	17.3%	33,530 KB	45,744 KB
Stress	9.17%	24.6%	42,382 KB	64,304 KB
Spike	3.72%	18.0%	34,357 KB	63,664 KB

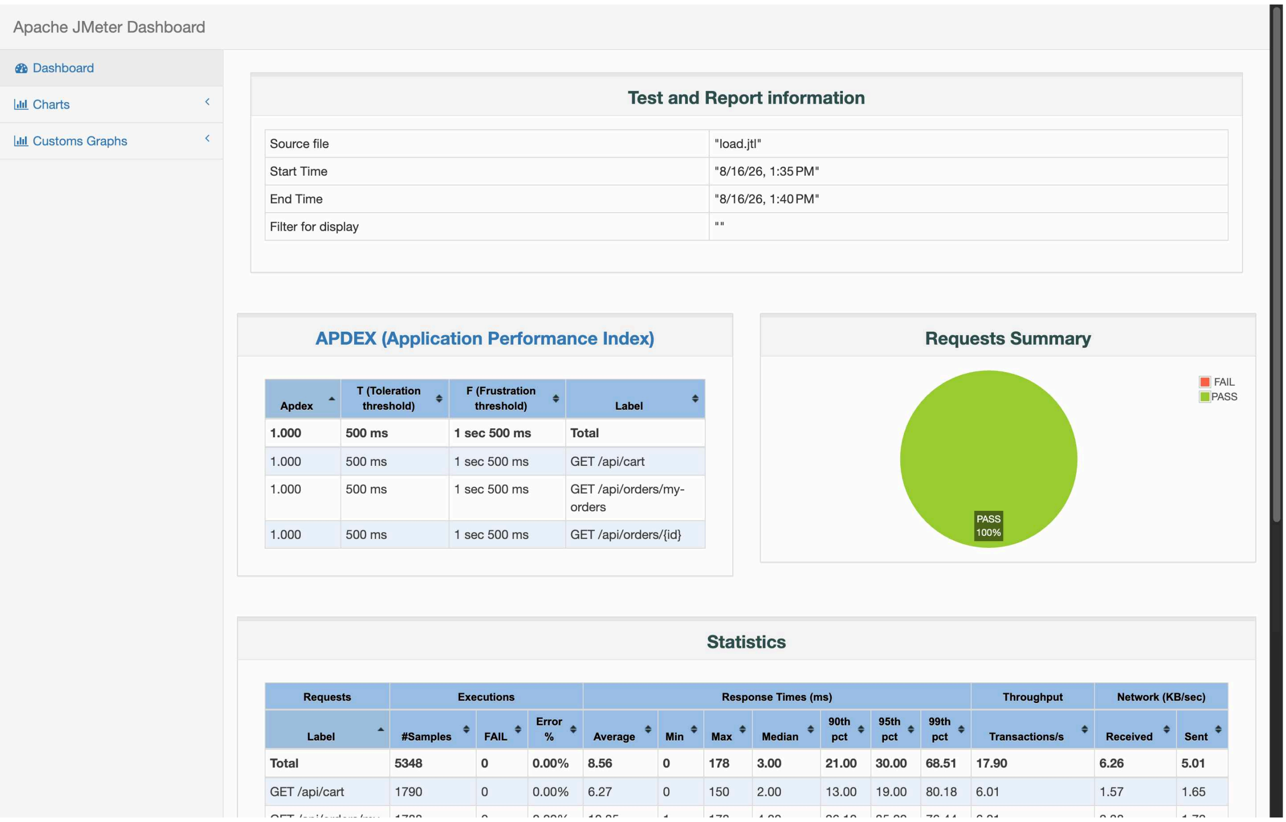
Stress used the most average CPU because registration and password reset write to SQLite. No measured run crashed, and all runs had a 0.00% error rate. I did not find a performance issue from these runs.

The following Load-run screenshot shows `htop` and the JMeter CLI output in the same frame. `htop` is filtered to the backend process, while JMeter shows active users and 0.00% errors during the captured period.



7.3 HTML dashboard evidence

I opened the generated JMeter dashboards in Chrome. These images show the rendered HTML reports from the raw JTL files.



Apache JMeter Dashboard

Dashboard

Charts

Customs Graphs

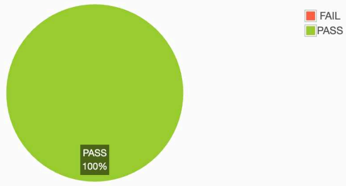
Test and Report information

Source file	"stress.jtl"
Start Time	"8/16/26, 1:41 PM"
End Time	"8/16/26, 1:44 PM"
Filter for display	""

APDEX (Application Performance Index)

Apdex	T (Toleration threshold)	F (Frustration threshold)	Label
0.998	**500 ms**	**1 sec 500 ms**	**Total**
0.997	500 ms	1 sec 500 ms	POST /api/forgot-password
0.999	500 ms	1 sec 500 ms	POST /api/reset-password
1.000	500 ms	1 sec 500 ms	POST /api/register

Requests Summary



Statistics

Requests	Executions			Response Times (ms)							Throughput	Network (KB/sec)	
Label	#Samples	FAIL	Error %	Average	Min	Max	Median	90th pct	95th pct	99th pct	Transactions/s	Received	Sent
Total	**8522**	**0**	**0.00%**	**25.06**	**1**	**648**	**5.00**	**56.00**	**139.00**	**339.00**	**47.53**	**15.11**	**12.20**
POST	2843	0	0.00%	29.53	1	648	6.00	66.00	158.40	431.56	15.93	5.44	3.60

Apache JMeter Dashboard

Dashboard

Charts

Customs Graphs

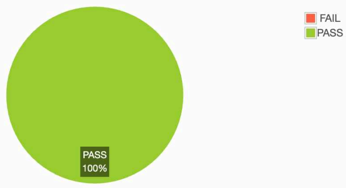
Test and Report information

Source file	"spike_combined.jtl"
Start Time	"8/16/26, 1:46 PM"
End Time	"8/16/26, 1:49 PM"
Filter for display	""

APDEX (Application Performance Index)

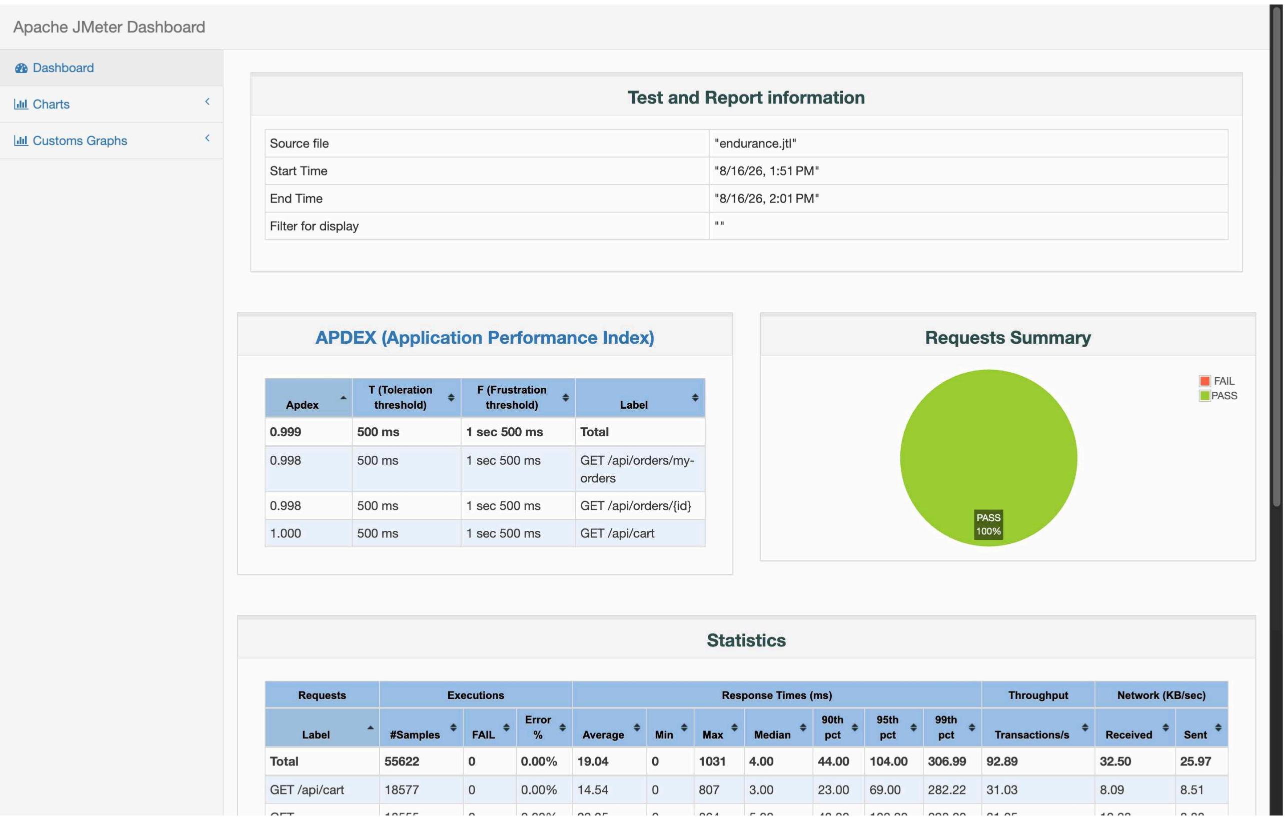
Apdex	T (Toleration threshold)	F (Frustration threshold)	Label
0.998	**500 ms**	**1 sec 500 ms**	**Total**
0.998	500 ms	1 sec 500 ms	POST /api/apply-coupon
0.998	500 ms	1 sec 500 ms	POST /api/admin/coupons
0.999	500 ms	1 sec 500 ms	DELETE /api/admin/coupons/{id}

Requests Summary



Statistics

Requests	Executions			Response Times (ms)							Throughput	Network (KB/sec)	
Label	#Samples	FAIL	Error %	Average	Min	Max	Median	90th pct	95th pct	99th pct	Transactions/s	Received	Sent
Total	**3711**	**0**	**0.00%**	**22.05**	**0**	**622**	**5.00**	**34.00**	**97.00**	**403.00**	**18.77**	**6.09**	**6.50**
DELETE /api/admin/coupons/{id}	1225	0	0.00%	21.21	1	514	5.00	34.40	84.10	407.22	6.27	1.81	2.14



8. Step 6 — Determine the endurance threshold

The endurance test ran for ten minutes with 100 users, a 60-second ramp-up, and a one-second think time. It completed 55,622 samples. The measured results were:

- throughput: **92.70 RPS** using the planned 600-second duration;
- mean response time: **19.04 ms**;
- p95 response time: **83 ms**;
- maximum response time: **1,031 ms**;
- errors: **0.00%**;
- backend peak CPU: **26.9%**; and
- backend peak RSS: **76,000 KB**.

Therefore, the tested stable level on this computer is **at least 92.70 RPS for ten minutes at 100 users**. This is not the absolute maximum capacity. The backend still had CPU headroom, and I did not increase traffic until failure.

9. Step 7 — Check the AI result analysis

After the tests, I asked the AI to analyse the raw JTL and resource CSV files. I then recalculated the important values.

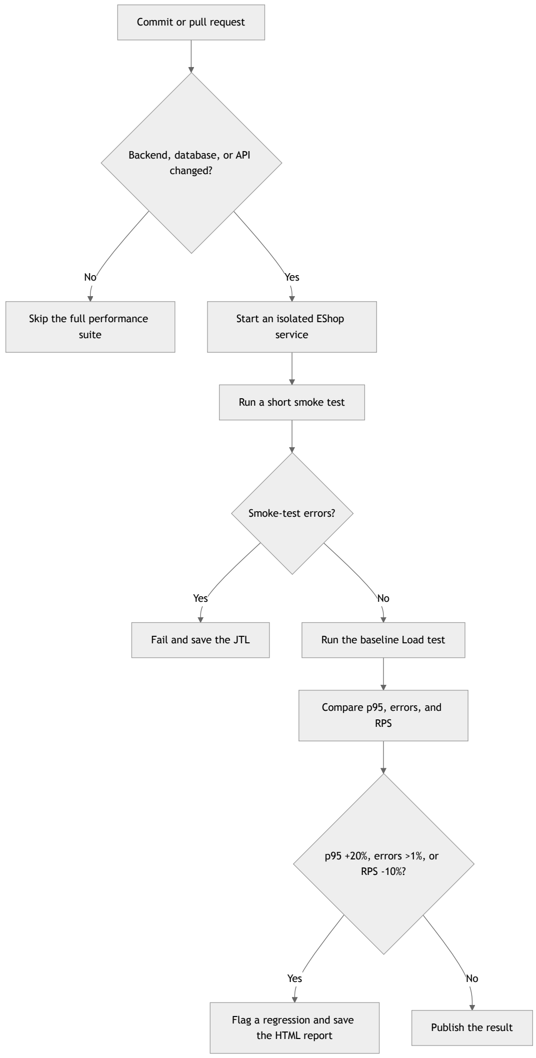
The AI reported a merged Spike p95 of 97 ms and a high-load-stage p95 of 105 ms. My direct check of the raw data gave **93 ms** for the merged JTL and **103 ms** for the high-load stage. The difference probably came from the percentile position or from mixing the merged and stage files.

The AI calculated endurance throughput as 92.89 RPS from the observed time window. I report 92.70 RPS because I divide 55,622 samples by the planned 600 seconds. Both calculations describe the same stable run, but neither proves an absolute hardware limit.

9.1 Review of AI recommendations

AI recommendation	My decision	Reason
Add an index on `orders(user_id, id)`	Feasible experiment	The order-history query filters by `user_id` and sorts by ID. Measure before and after.
Add an index on `users(email)`	Feasible experiment	Register and password-reset operations use email lookup.
Enable SQLite WAL	Feasible experiment	It may improve concurrent writes, but the current results do not prove that it is needed.
Add a normal database connection pool	Unsuitable	This SUT uses one local SQLite connection, not a remote database server.
Add caching now	Not supported by evidence	CPU was low and there were no errors. A slow query should be identified first.

10. Step 8 — Continuous performance testing proposal



The short smoke and Load tests should run when a pull request changes backend, database, or API code. Stress and endurance tests should run nightly or before a release because they take more time.

There are two main trade-offs. First, long tests use more CI time and money. Second, a busy shared runner may create a false alarm. To reduce false alarms, the pipeline should use fixed test data and the same runner type. It should also repeat a failed performance comparison before blocking a release.

11. Conclusion

I completed the Load, Stress, Spike, and endurance runs with raw JTL files, resource CSV files, JMeter logs, HTML reports, and dashboard screenshots. All measured runs had 0.00% errors. The strongest tested stable result was 92.70 RPS for ten minutes with 100 users, p95 83 ms, and backend peak RSS 76,000 KB.

The AI helped create a first plan and analyse the results, but it did not own the final answer. I changed the plan for the local environment and checked the important metrics from the raw files. The separate 200–300 word critique is in `AI_Critique.md`. The AI interaction record is in `AI_Audit_Report.md`.